

Core Model Proposal #403:

Miscellaneous Bugfixes Prior to 2025 Annual Meeting

Product: Global Change Analysis Model (GCAM)

Institution: Joint Global Change Research Institute (JGCRI)

Authors: Pralit Patel, Patrick O'Rourke, Xin Zhao, Maridee Weber

Reviewers: Marshall Wise, Gokul Iyer

Date committed: 6/2/2025

IR document number: PNNL-37802

Related sector: Energy, Land, Emissions

Type of development: Bugfix

Purpose: Miscellaneous bugfixes.

Purpose

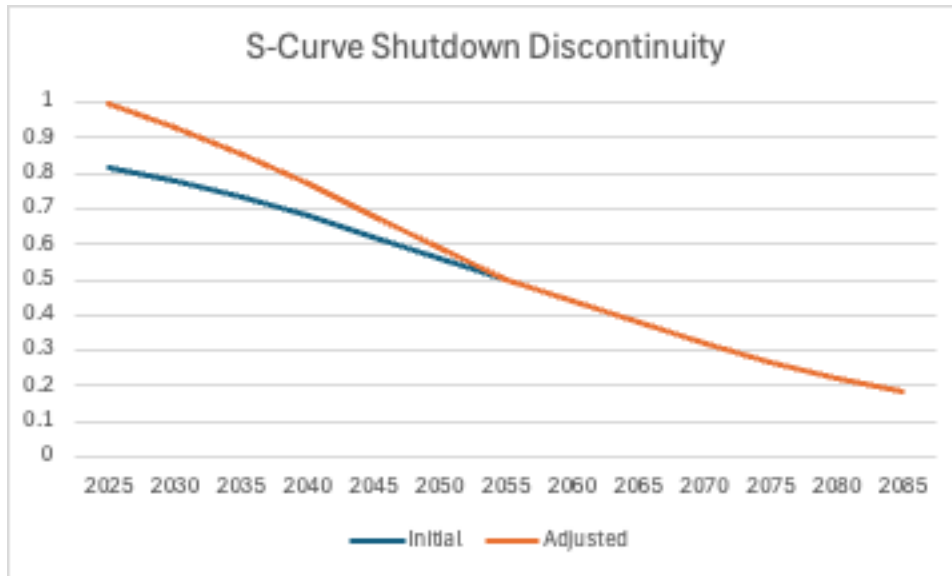
This core model proposal contains various bugfixes from GCAM v7.1+ to resolve various minor issues and improve model behavior.

Description of Changes

Resolve discontinuity in S-Curve shutdown decider

The the S-Curve shutdown decider is a type of "shutdown" function which smoothly decays a vintage over several model periods. However, the functional form that we use asymptotically reaches when (no shutdown) at negative infinity, not zero. What this means practically is that there is a bit of a "cliff" in the first vintaged year of a technology, which is particularly noticeable at less steep parameterizations.

To address this issue we adjust the formulation by calculating how far off from 1 the parametrization would be at zero year and linearly phase out that difference by the configured half life. In this way there is no initial "cliff" and behavior converges to the un-adjusted formulation.



A comparison of the initial formulation, parameterized with a lower than typical steepness value to highlight the issue. In this parameterization a vintage built in 2025 would currently retire about 17% of its stock after 5 year compared to just 7% with the new formulation which is generally the behavior users are looking for.

Fix transportation technology names for non-CO2 emissions

Several road-based transportation technologies did not have improvement or minimum coefficient assumptions for their non-CO2 emission factors. This is because they had their technology names revised in a previous release of GCAM but the revised technology names were missing from the relevant non-CO2 input files within GCAM's data system. Technology names have been corrected here for the following technologies:

- Light truck (liquids and natural gas)
- Medium truck (liquids and natural gas)
- Heavy truck (liquids and natural gas)
- 2W and 3W (liquids and natural gas)
- Car (liquids, hybrid liquids, and natural gas)
- Large car and truck (liquids, hybrid liquids, and natural gas)

Minimum coefficient assumptions by emission species for the above technologies are in Table 1. The "steepness", or percent improvement, for each of these technologies is 3.5% for all relevant non-CO2 emission species.

Technology group	SO2	NOx	CO	BC	OC	NMVOC
Liquids	0.0005	0.0054	0.0546	0.0016	0.0037	0.0065
Hybrid liquids	0.0005	0.0054	0.0546	0.0016	0.0037	0.0065
Natural gas	0.0033	0.2725	1.8195	0	0	0.0648

Table 1. Minimum coefficient by emission species and transportation technology group (kg/GJ).

Update land logit exponent

In the CMP #393 (Zhao et al. 2024), the land logit exponent in the agroforest_nonpasture nest was supposed to be changed from 1.25 to 0.75 (see the Figure below). But that change was missed. This is fixed now. The conversion between forest and cropland will be more rigid given the change.

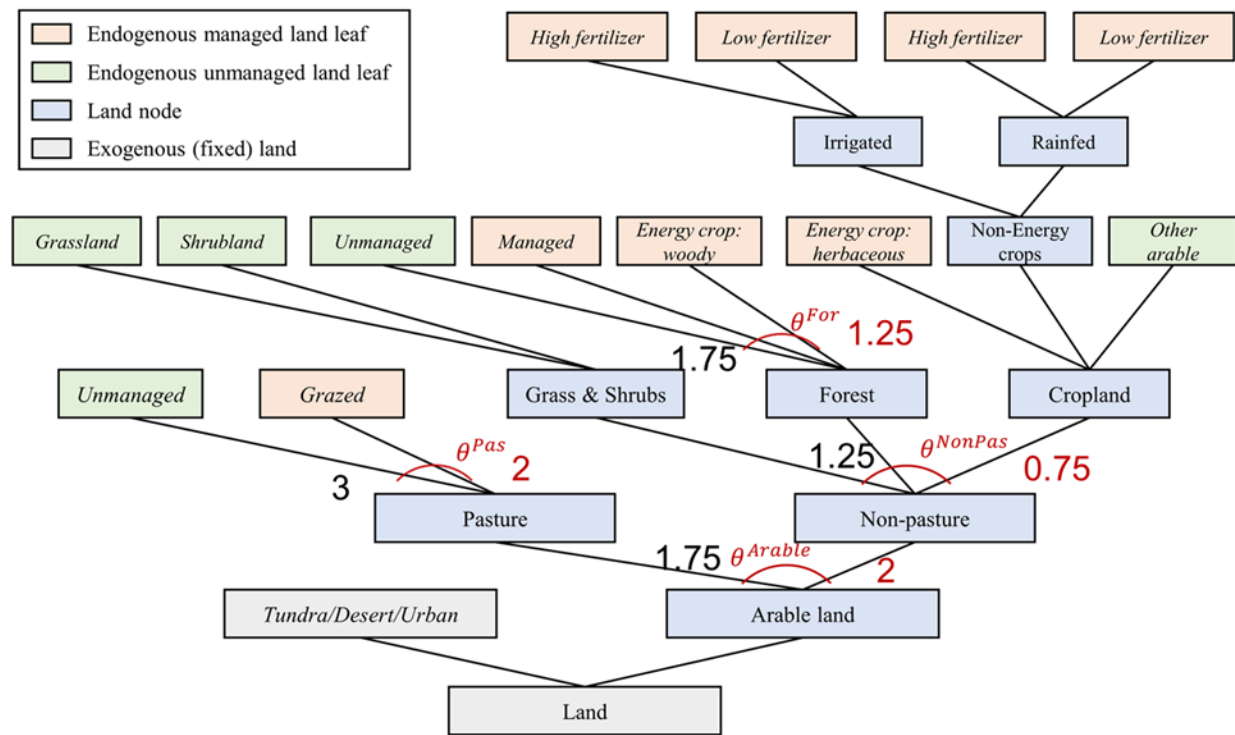


Fig. 1 Land nesting structure in GCAM (Fig. 1 from CMP #393). The set of key logit exponent parameters proposed was (valued in red) $\theta^{Arable}=1.75$, $\theta^{Pas}=2$, $\theta^{NonPas}=0.75$, and $\theta^{For}=1.25$ (new values in red and original values in black).

Fix a GCAM-USA bug that results in no f-gas emissions in the residential sector

A bug in `zgcamusa_L273.en_ghg_emissions` resulted in no f-gas emissions being assigned to residential buildings technologies. This bug was caused by a joining error introduced by the implementation of multiple consumers/income deciles in [CMP #362](#). In addition to this bugfix, the methodology for allocating f-gas emissions to different technologies was improved. The updated methodology uses the [U.S. GHG Inventory](#) (Table 4-127: Emissions of HFCs, PFCs, and CO₂ from ODS Substitutes (MMT CO₂ Eq.) by Sector) to share out emissions by technology (commercial/residential air conditioning/refrigeration) over time, and then assigns them to the U.S. states based on electricity consumption from these technologies.

Table 4-127: Emissions of HFCs, PFCs, and CO₂ from ODS Substitutes (MMT CO₂ Eq.) by Sector

Sector	1990	2005	2018	2019	2020	2021	2022
Refrigeration/Air Conditioning	+	83.0	122.7	126.5	130.6	139.5	144.6
Commercial Refrigeration	+	14.9	39.6	40.2	40.6	41.0	41.4
Domestic Refrigeration	+	0.2	1.2	1.2	1.2	1.1	1.0
Industrial Process Refrigeration	+	1.8	13.8	15.0	16.2	17.4	18.6
Transport Refrigeration	+	1.6	6.9	7.4	7.9	8.4	8.8
Mobile Air Conditioning	+	61.5	28.7	26.6	24.6	22.9	20.8
Residential Stationary Air Conditioning	+	1.2	26.2	29.4	33.2	41.5	46.4
Commercial Stationary Air Conditioning	+	1.7	6.2	6.6	6.9	7.3	7.6
Aerosols	0.2	10.2	16.7	17.0	17.3	17.7	17.0
Foams	+	3.5	14.2	14.1	13.7	10.8	11.7
Solvents	+	1.6	2.0	2.0	2.0	2.1	2.1
Fire Protection	+	1.2	2.4	2.5	2.5	2.6	2.6
Total	0.3	99.5	157.9	162.1	166.2	172.7	178.1

+ Does not exceed 0.05 MMT CO₂ Eq.

Note: Totals may not sum due to independent rounding.

Fig. 2 Emissions of HFCs, PFCs, and CO₂ as ozone depleting substance substitutes by end-use sector for 1990 through 2022.

Before this bugfix (core), f-gas emissions from residential buildings were being accidentally dropped. After fixing this (branch), emissions from f-gasses increase since all emissions are now accounted for.

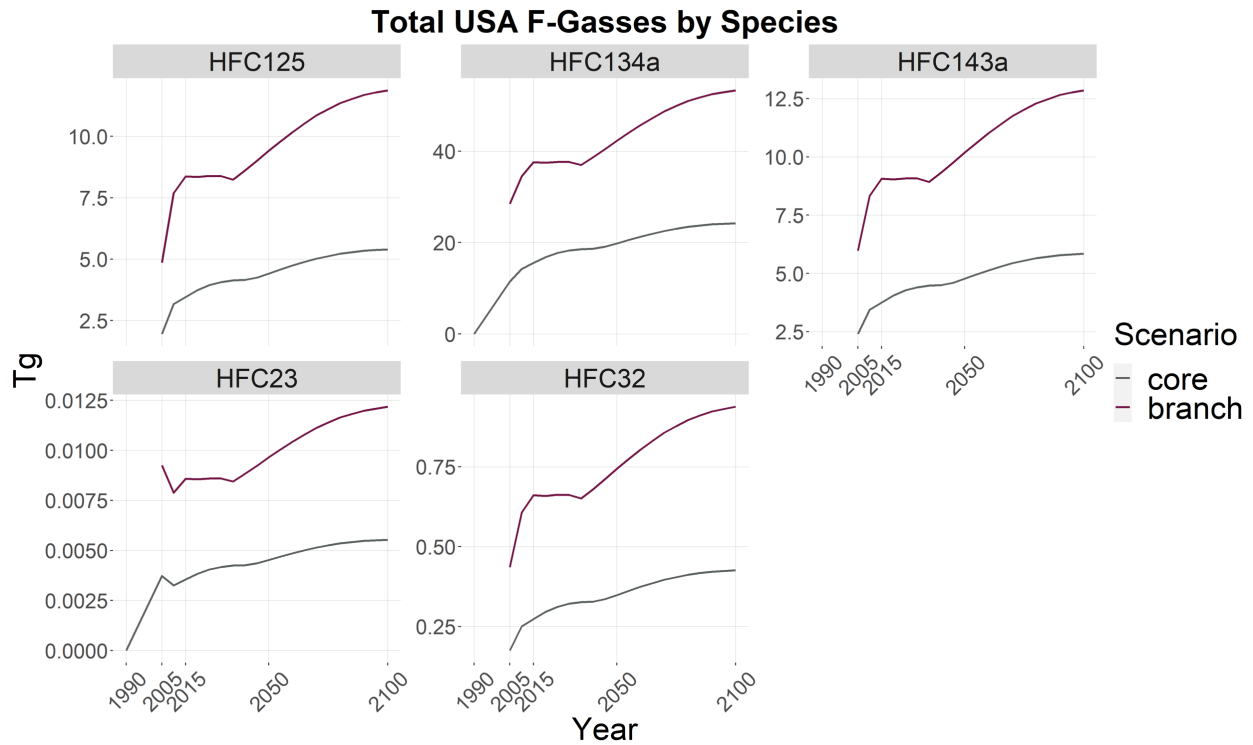


Fig. 3 Aggregated GCAM-USA state-level f-gas emissions by species.

Additionally, the bugfix and improved methodology result in emissions for f-gasses for multiple technologies. Previously (core), emissions only existed for commercial cooling. Now (branch), we have emissions for commercial cooling and refrigeration, and residential cooling and refrigerators. Note that f-gas emissions begin in 2005 on this branch, as the US GHG Inventory showed zero or near-zero emissions for these technologies in 1990.

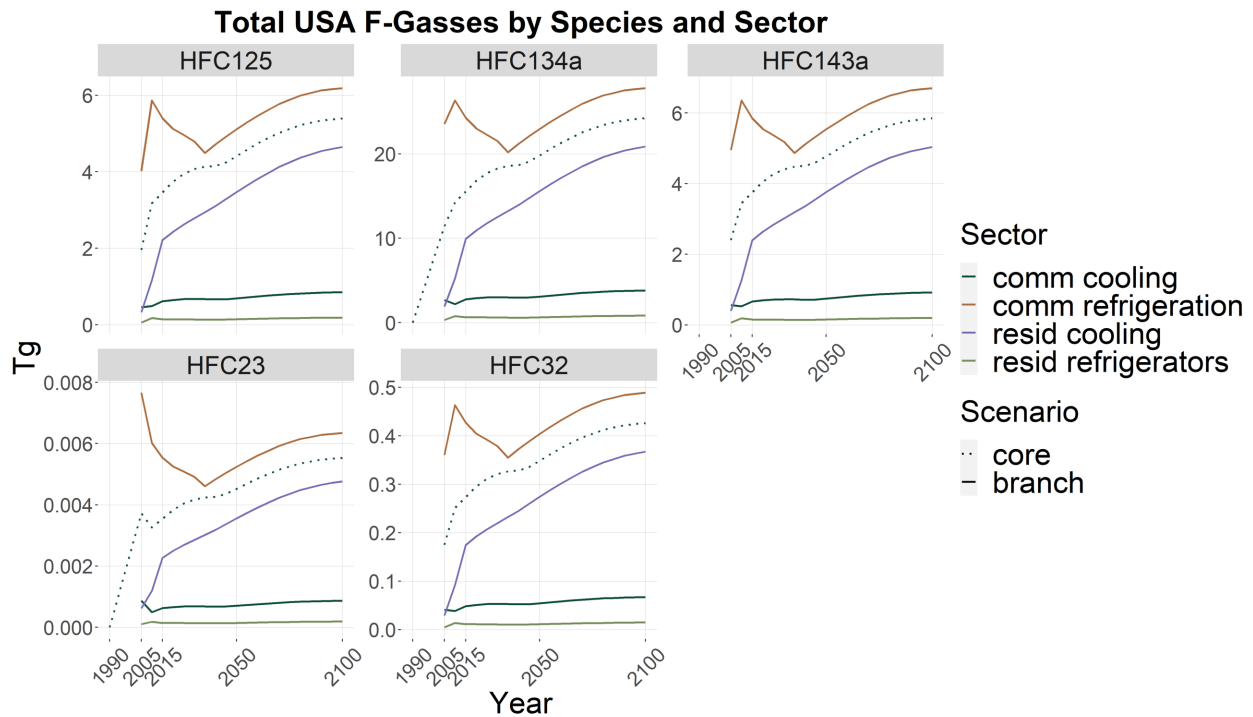
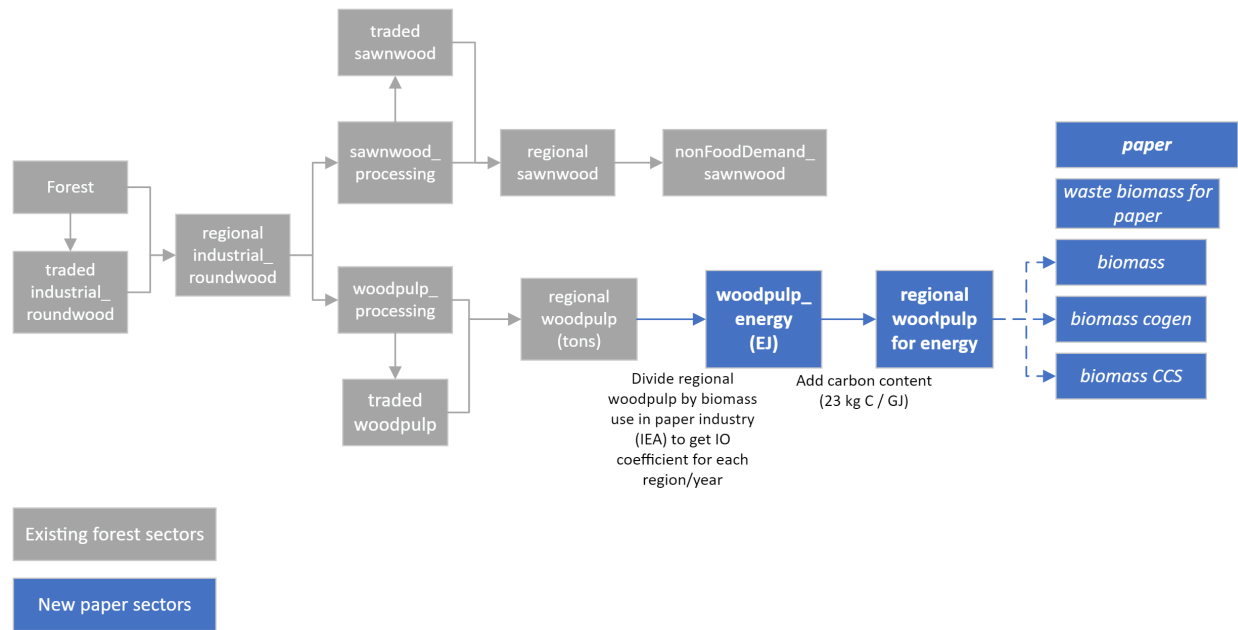


Fig. 4 Aggregated GCAM-USA state-level f-gas emissions by species and sector/technology.

Fix Unintended CCS capture of Paper products

Given the large regional variability in the ratio of wood pulp going into the paper industry that is used for feedstock vs energy we collapse the two flows into a singular woodpulp_energy commodity in GCAM. When used without CCS both the carbon in the energy and feedstock are emitted immediately, consistent with the way we treat other Forestry products. However, if CCS is in use it can capture the full carbon amount, including the feedstock amount which was not intended.

To address this issue we simply adjust the remove fraction of the CCS to only remove the carbon in the energy portion of the woodpulp_energy commodity.



Schematic from the original CMP 370 highlighting the flow of wood pulp into the paper industry and its treatment wrt biomass energy and CCS.

Update Superwell data to include choice of depletion scenario

Update the raw Superwell data to a version which includes multiple depletion scenarios which users can choose from by updating the `water.GROUNDWATER_SCENARIO` constant. Note this feature existed previously but was lost in the last Superwell data update.

Remove unused C++ code

WindTechnology and SolarTechnology classes were removed from the C++ code and project files.

Various fixes to pass gcamdata tests

We update numerous chunks to include fixes such as:

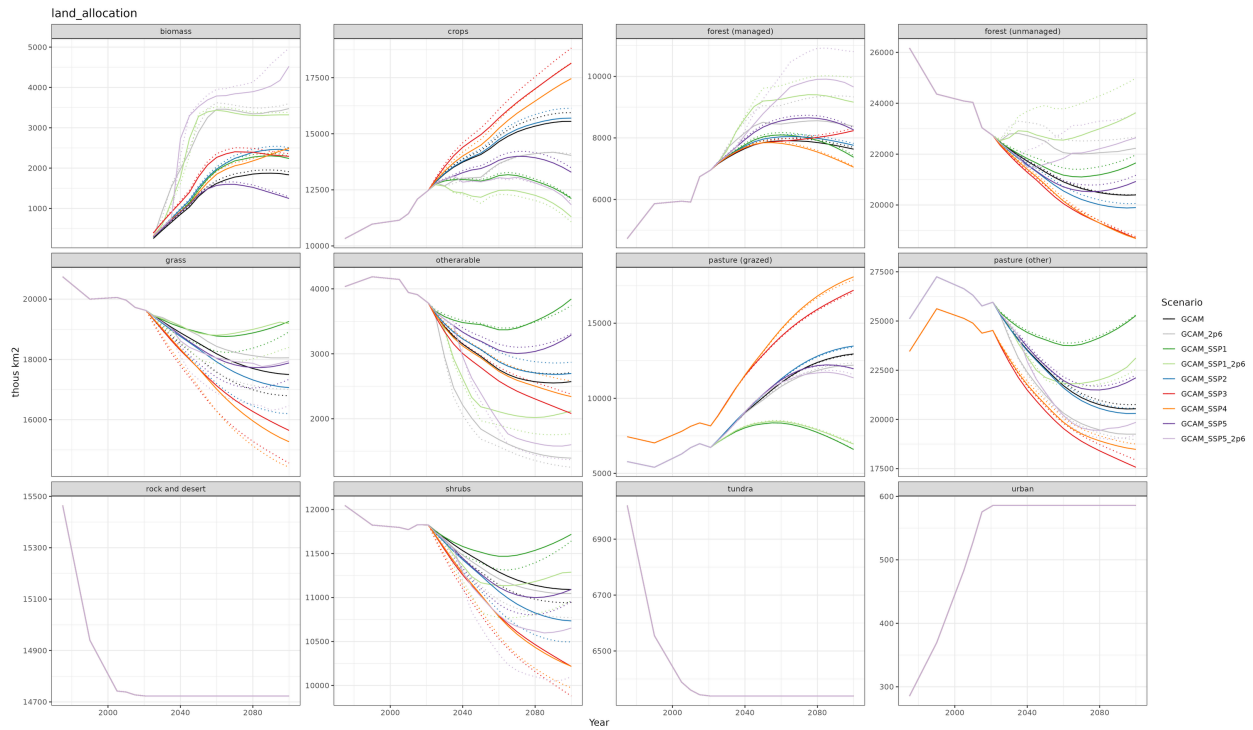
- All capitalized constants
- No consecutive `mutate()` calls (combine into a single operation)
- Use `if_else()` instead of `ifelse()`
- Use `dplyr` joins instead of base R `merge()` function
- Ensure all precursors are appropriately referenced

Avoid warnings related to old Backup Electricity tables

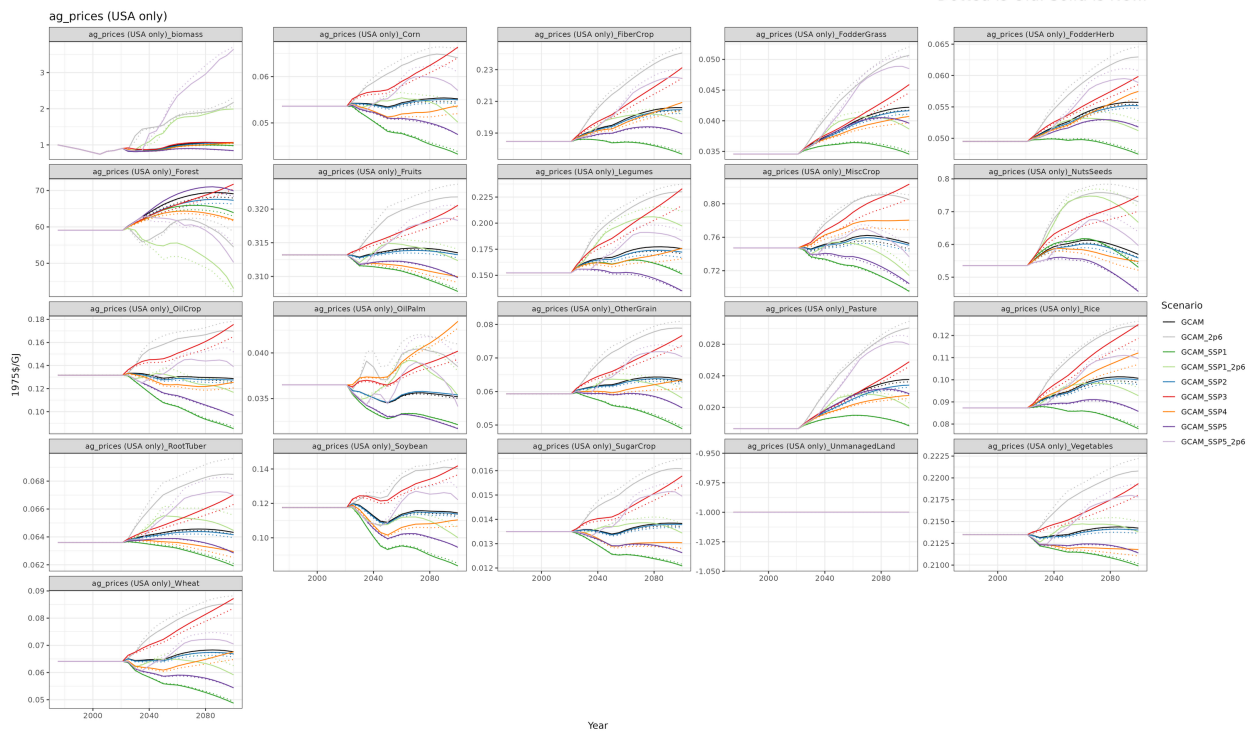
Add additional checks to avoid generating warnings in gcamdata as well as in GCAM when energy.ELEC_USE_BACKUP is false.

Validation

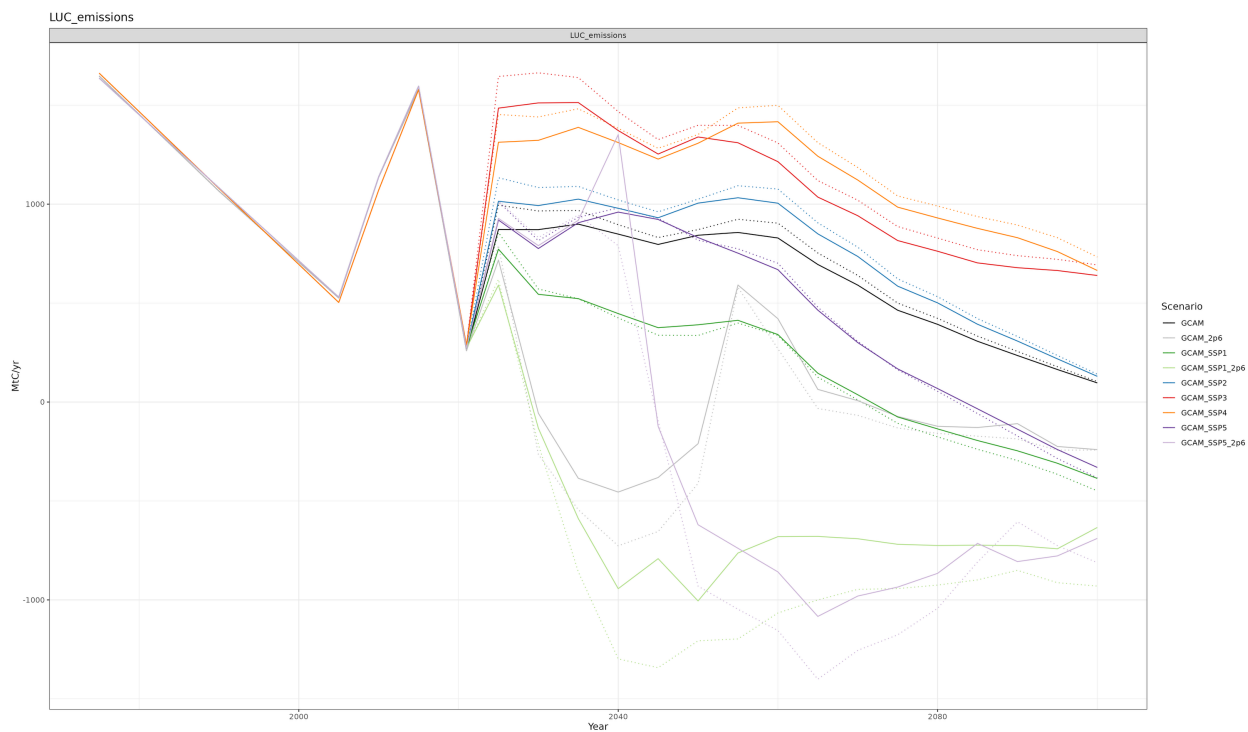
The fix which has the most impact, particularly in 2.6 scenarios is the Logit exponent reduction which reduces the substitution between Forest land and other types of land such as Crops. For 2.6 scenarios that have terrestrial carbon prices this implies we should expect less mitigation through afforestation while at the same time limiting price increases for Crop commodities in those scenarios. The lack of mitigation through afforestation implies more mitigation needs to occur in the energy system.



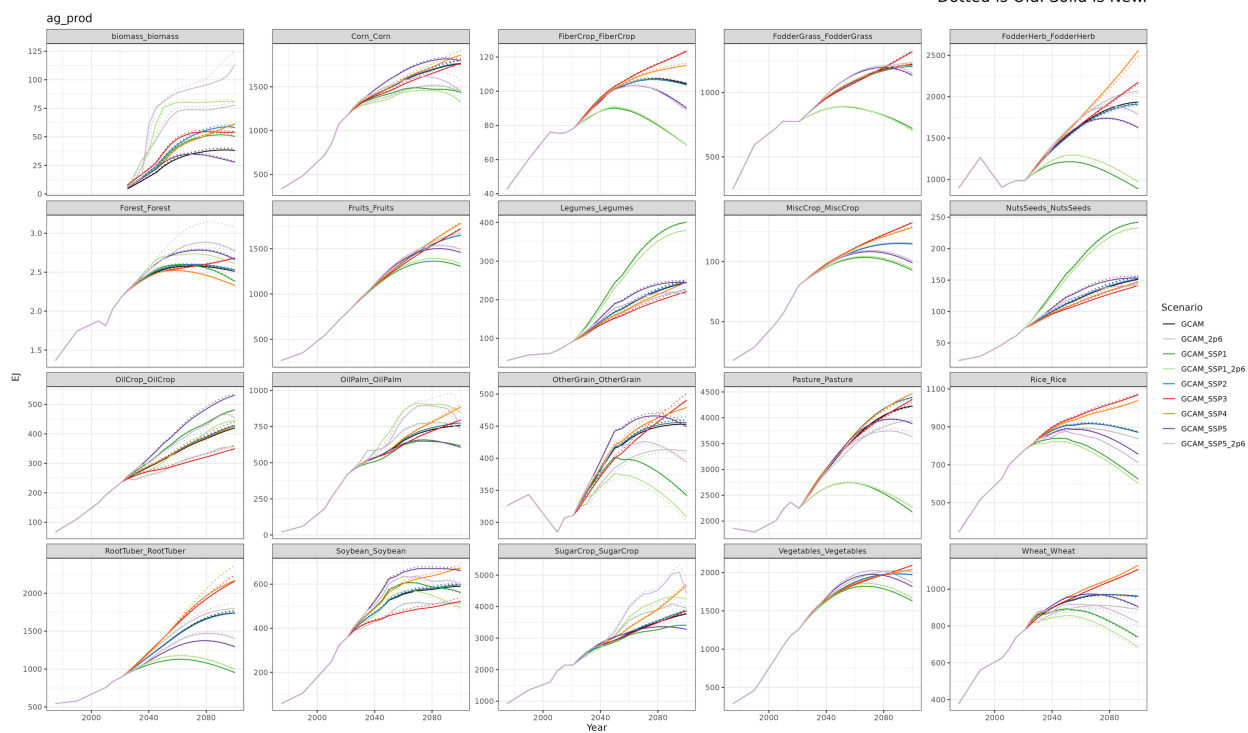
Dotted is Old. Solid is New.



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